Lecture 17

Unitary induction

Objectives. *Lecture 12 constructed induced representations of compact groups using an invariant measure on K/H. For a noncompact group G, the quotient G/H may carry no invariant measure. We replace it by a* quasi-invariant *measure, which translation by an element of G multiplies by a positive continuous function. We then correct the action by a square-root factor. The model is the $ax+b$ group, with representations induced from characters of its translation subgroup.*

By the end of this lecture you will be able to:

- State the quotient-measure theorem and derive the quasi-invariance factor and its cocycle identity.
- Define the normalised induced representation and prove that it is unitary.
- Compute the $ax+b$ representations induced from translation characters in two measure pictures.
- Explain why the unitary equivalence class of the induced representation depends only on the scaling orbit of the frequency.

17.1 Equivariant functions

Throughout, G is a Lie group, H is a closed subgroup and $q(x) = xH$ is the quotient map $G \to G/H$. By Lecture 3, G/H is a manifold on which G acts by $g \cdot xH = gxH$. The map q is continuous and open. Let (σ, V) be a unitary representation of H on a Hilbert space V.

Definition 17.1. An *equivariant function of type σ* is a continuous map $f : G \to V$ with

$$f(xh) = \sigma(h)^{-1} f(x) \qquad (x \in G,\ h \in H). \tag{17.1}$$

Let $\mathcal{F}_c(\sigma)$ be the space of those f for which $q(\operatorname{supp} f)$ is compact.

This is the convention of Lecture 12. Let $s : U \to G$ be a continuous section of q over an open set $U \subseteq G/H$. Then $f(s(p)h) = \sigma(h)^{-1} f(s(p))$. So over U the function f is just the ordinary function $f \circ s : U \to V$. Continuous local sections exist near every point; they are part of the construction of G/H in Lecture 3. Geometrically f is a section of a vector bundle over G/H. We use only the function description.

Since $\sigma(h)$ is unitary, $n_f(xH) = \|f(x)\|^2$ is well defined. It is continuous on G/H because q is open, and its support is compact.

Averaging over H turns compactly supported continuous functions $G \to V$ into elements of $\mathcal{F}_c(\sigma)$. Fix a left Haar measure dh on H. For continuous compactly supported $F\colon G \to V$ define $P_\sigma F$ by

$$P_\sigma F(x) = \int_H \sigma(h) F(xh)\, dh.$$

The integral is defined weakly by the Riesz argument of Lecture 10, applied to a continuous Hilbert-space-valued function with compact support. For fixed x, the integrand vanishes outside $H \cap x^{-1} \operatorname{supp} F$.

Left invariance of dh under $h' = h_0 h$ gives

$$P_\sigma F(xh_0) = \int_H \sigma(h_0^{-1}h') F(xh')\, dh' = \sigma(h_0)^{-1} P_\sigma F(x).$$

For x in a compact neighbourhood, all the integrands have support in one compact subset of H. Uniform continuity there makes $P_\sigma F$ continuous. Also $q(\operatorname{supp} P_\sigma F) \subseteq q(\operatorname{supp} F)$, which is compact. Thus $P_\sigma F \in \mathcal{F}_c(\sigma)$.

17.2 Quotient measures

As in Lecture 16, a *measure* on a locally compact space X here means a positive linear functional $\varphi \mapsto \int_X \varphi\, d\mu$ on $C_c(X)$ which is strictly positive on nonzero $\varphi \geq 0$. Then $L^2(X, \mu)$ is the completion of $C_c(X)$ for $\langle \varphi, \psi \rangle = \int_X \bar{\varphi}\psi\, d\mu$.

Fix left Haar measures dx on G and dh on H. Modular functions follow Lecture 16:

$$\int_G F(xg)\, dx = \Delta_G(g)^{-1} \int_G F(x)\, dx,$$

and likewise for Δ_H. For $F \in C_c(G)$ define $F^H \in C_c(G/H)$ by

$$F^H(xH) = \int_H F(xh)\, dh. \tag{17.2}$$

It is well defined by left invariance of dh.

Lemma 17.2. *If a measure μ on G/H satisfies $\int_G F\, dx = \int_{G/H} F^H\, d\mu$ for all $F \in C_c(G)$, then $\Delta_G(h) = \Delta_H(h)$ for all $h \in H$.*

Proof. Fix $h_0 \in H$ and apply the identity to $F_{h_0}(x) = F(xh_0)$. The left side becomes $\Delta_G(h_0)^{-1} \int_G F\, dx$. On the right,

$$(F_{h_0})^H(xH) = \int_H F(xhh_0)\, dh = \Delta_H(h_0)^{-1} F^H(xH).$$

Hence $\Delta_G(h_0)^{-1} \int_G F\, dx = \Delta_H(h_0)^{-1} \int_G F\, dx$. Choose $F \geq 0$ nonzero, so that $\int_G F\, dx > 0$. Dividing both sides by $\int_G F\, dx$ gives $\Delta_G(h_0)^{-1} = \Delta_H(h_0)^{-1}$, so $\Delta_G(h_0) = \Delta_H(h_0)$. ∎

For the $ax + b$ group and its scaling subgroup A, we will see $\Delta_G(a, 0) = 1/a$ while $\Delta_A = 1$. Hence, by the modular compatibility lemma, Lemma 17.2, G/A carries no such measure. Instead, we insert a weight function on G into the integral over G.

Theorem 17.3 (Quotient measures; cited)**.** *Let G be a Lie group and H a closed subgroup.*

(a) There is a continuous function $\rho\colon G \to (0,\infty)$ with

$$\rho(xh) = \frac{\Delta_H(h)}{\Delta_G(h)}\,\rho(x) \qquad (x \in G,\ h \in H). \tag{17.3}$$

(b) The map $F \mapsto F^H$ sends $C_c(G)$ onto $C_c(G/H)$. If $\varphi \geq 0$, a lift F with $F^H = \varphi$ can be chosen with $F \geq 0$.

(c) For each ρ as in (a) there is a unique measure μ_ρ on G/H with

$$\int_G F(x)\rho(x)\,dx = \int_{G/H} F^H\,d\mu_\rho \qquad (F \in C_c(G)). \tag{17.4}$$

For a general closed subgroup, see G. B. Folland, *A Course in Abstract Harmonic Analysis*, second edition (2016), Chapter 2, “Measures on Homogeneous Spaces”, cited below as Folland, Abstract Harmonic Analysis. Its modular convention agrees with ours. Uniqueness of μ_ρ follows from the onto property of the averaging map. A. A. Kirillov, *Elements of the Theory of Representations* (1976), Section 9.1, cited below as Kirillov, Elements, treats quasi-invariant measures and the modular criterion. Its conventions must be translated before use.

A continuous positive function ρ with the transformation law (17.3) is called a *rho-function*. Repeating the computation in the modular compatibility lemma, Lemma 17.2, with $F\rho$ in place of F confirms that the transformation law (17.3) is compatible with our modular convention. If $\Delta_G|_H = \Delta_H$ we may take $\rho = 1$.

Proposition 17.4. *Let ρ be a rho-function and $\mu = \mu_\rho$. Define λ by $\lambda(g, xH) = \rho(gx)/\rho(x)$. Then λ is well defined. For $g, g_1, g_2 \in G$, $p \in G/H$ and $\varphi \in C_c(G/H)$,*

$$\lambda(g_1g_2, p) = \lambda(g_1, g_2p)\,\lambda(g_2, p), \qquad \lambda(e, p) = 1, \tag{17.5}$$

$$\int_{G/H} \varphi(g^{-1}p)\,d\mu(p) = \int_{G/H} \varphi(p)\,\lambda(g, p)\,d\mu(p). \tag{17.6}$$

Proof. By the transformation law (17.3), replacing x by xh multiplies $\rho(gx)$ and $\rho(x)$ by the same factor. So λ is well defined. The function $(g, x) \mapsto \rho(gx)/\rho(x)$ is continuous and constant on fibres of $\mathrm{id} \times q$. Since q is open, $\mathrm{id} \times q$ is an open surjection, hence a quotient map. Thus λ is jointly continuous in (g, p). For $p = xH$,

$$\lambda(g_1g_2, p) = \frac{\rho(g_1g_2x)}{\rho(g_2x)} \cdot \frac{\rho(g_2x)}{\rho(x)} = \lambda(g_1, g_2p)\,\lambda(g_2, p).$$

Also $\lambda(e, p) = \rho(x)/\rho(x) = 1$.

For the quasi-invariance identity (17.6), use the onto property of averaging, Theorem 17.3(b), to write $\varphi = F^H$ with $F \in C_c(G)$. The function $x \mapsto F(g^{-1}x)$ has average $\varphi(g^{-1}\,\cdot)$. By the weighted quotient integral (17.4) and left invariance of dx, with $y = g^{-1}x$,

$$\int \varphi(g^{-1}p)\,d\mu(p) = \int_G F(g^{-1}x)\rho(x)\,dx = \int_G F(y)\rho(gy)\,dy = \int_G F(y)\,\lambda(g, yH)\,\rho(y)\,dy.$$

Since $\lambda(g, \cdot)$ is constant on cosets, $y \mapsto F(y)\lambda(g, yH)$ has average $\varphi\,\lambda(g, \cdot)$. Applying the weighted quotient integral (17.4) again gives the right side of (17.6). ∎

We say μ is *quasi-invariant*: translation by g changes it by the positive continuous factor $\lambda(g,\cdot)$. If $\rho = 1$, then $\lambda = 1$ and μ is G-invariant.

17.3 The normalised induced representation

Fix a rho-function ρ, and let μ and λ be as in the quasi-invariance proposition, Proposition 17.4. For $f_1, f_2 \in \mathcal{F}_c(\sigma)$ put

$$\langle f_1, f_2 \rangle = \int_{G/H} \langle f_1(x), f_2(x) \rangle \, d\mu(xH), \qquad \|f\|^2 = \int_{G/H} n_f \, d\mu.$$

The integrand is well defined because σ is unitary. If $\|f\| = 0$, then $n_f = 0$ by strict positivity, so $f = 0$.

Theorem 17.5. *For $f \in \mathcal{F}_c(\sigma)$ and $g \in G$ define*

$$(\Pi(g)f)(x) = \left(\frac{\rho(g^{-1}x)}{\rho(x)}\right)^{1/2} f(g^{-1}x) = \lambda(g^{-1}, xH)^{1/2} f(g^{-1}x). \tag{17.7}$$

Then (a) $\Pi(g)f \in \mathcal{F}_c(\sigma)$; (b) $\Pi(e) = I$ *and* $\Pi(g_1 g_2) = \Pi(g_1)\Pi(g_2)$; (c) $\|\Pi(g)f\| = \|f\|$.

Proof. (a) The factor depends only on xH. So replacing x by xh only changes $f(g^{-1}x)$ into $\sigma(h)^{-1} f(g^{-1}x)$. Also $q(\operatorname{supp} \Pi(g)f) = g \cdot q(\operatorname{supp} f)$ is compact.

(b) Let $p = xH$. Applying the normalised action (17.7) twice,

$$(\Pi(g_1)\Pi(g_2)f)(x) = \lambda(g_1^{-1}, p)^{1/2} \, \lambda(g_2^{-1}, g_1^{-1}p)^{1/2} \, f(g_2^{-1} g_1^{-1} x).$$

The cocycle identity for the pair g_2^{-1}, g_1^{-1} turns the product of factors into $\lambda((g_1 g_2)^{-1}, p)^{1/2}$. Also $\lambda(e, p) = 1$ gives $\Pi(e) = I$.

(c) Translation by g changes μ by the factor $\lambda(g,\cdot)$. The square-root factor in $\Pi(g)$ is chosen to cancel this change. We have $n_{\Pi(g)f}(p) = \lambda(g^{-1}, p)\, n_f(g^{-1}p)$. Put $\varphi(p) = n_f(p)\, \lambda(g^{-1}, gp)$, so that $\varphi(g^{-1}p) = n_{\Pi(g)f}(p)$. Then the quasi-invariance identity (17.6) gives

$$\|\Pi(g)f\|^2 = \int \varphi(g^{-1}p)\, d\mu = \int n_f(p)\, \lambda(g^{-1}, gp)\, \lambda(g, p)\, d\mu(p) = \int n_f \, d\mu.$$

The last step is the cocycle identity (17.5) with $g_1 = g^{-1}$ and $g_2 = g$. ∎

By the representation law and the isometry property in the normalised-induction theorem, Theorem 17.5, each $\Pi(g)$ is an isometry of $\mathcal{F}_c(\sigma)$ with inverse $\Pi(g^{-1})$. So $\Pi(g)$ extends to a unitary operator on the completion. The representation law persists by continuity.

Definition 17.6. The *normalised induced representation* $\operatorname{Ind}_H^G \sigma$ is the completion of $\mathcal{F}_c(\sigma)$, for the inner product defined by the chosen rho-function ρ, together with these unitary operators $\Pi(g)$.

Proposition 17.7. *The induced representation $\operatorname{Ind}_H^G \sigma$ is strongly continuous.*

Proof. First take $f \in \mathcal{F}_c(\sigma)$ and let $C = q(\operatorname{supp} f)$, a compact set. Choose a compact neighbourhood B of e in G. For $g \in B$, the quotient supports of both $\Pi(g)f$ and f lie in the compact set

$$S = B \cdot C \subset G/H.$$

It suffices to show that their difference tends uniformly to zero in fibre norm on S. This norm is independent of the representative in G because σ is unitary.

By compactness of S and the manifold structure of G/H, we can cover S by finitely many compact subsets S_i. Each S_i lies in an open set with a continuous local section s_i of q. The lifted sets $s_i(S_i)$ are compact. At these representatives,

$$(\Pi(g)f - f)(s_i(p)) = \lambda(g^{-1}, p)^{1/2} f(g^{-1}s_i(p)) - f(s_i(p)).$$

The function $(g, p) \mapsto \lambda(g^{-1}, p)$ is jointly continuous and equals one at $g = e$. Also $(g, p) \mapsto f(g^{-1}s_i(p))$ is jointly continuous as a Hilbert-space-valued function. On the compact sets $B \times S_i$, this continuity is uniform. Thus the displayed difference tends uniformly to zero for $p \in S_i$ as $g \to e$. There are only finitely many S_i, so the same holds on S.

The measure μ is finite on compact sets, by its Radon-measure interpretation in Lecture 16. So if the uniform bound is ε, then

$$\|\Pi(g)f - f\|^2 \le \varepsilon^2 \mu(S).$$

This proves norm continuity on $\mathcal{F}_c(\sigma)$.

For a vector v in the completion, choose f with $\|v - f\| < \varepsilon$. Unitarity gives

$$\|\Pi(g)v - v\| \le 2\varepsilon + \|\Pi(g)f - f\|.$$

Hence continuity holds at e for every v. By the representation law and unitarity, continuity at e implies continuity at every g_0. ∎

Remark 17.8. Without the square root, the map $f \mapsto f(g^{-1}\,\cdot)$ still preserves $\mathcal{F}_c(\sigma)$ and satisfies the representation law. By the quasi-invariance identity (17.6), this map sends $\|f\|^2$ to $\int n_f\, \lambda(g, \cdot)\, d\mu$. So in general it is not unitary; in the $ax+b$ example it multiplies $\|\psi\|^2$ by a_0 (Proposition 17.9(a)). For compact G and finite-dimensional σ, we take $\rho = 1$ and recover the construction of Lecture 12. Different rho-functions give unitarily equivalent representations (Exercise 17.3). Finite-group Frobenius reciprocity does not transfer verbatim, and we do not use it.

17.4 Representations of the $ax + b$ group induced from translations

Let $G = \{(a, b) : a > 0,\ b \in \mathbb{R}\}$ with

$$(a, b)(a', b') = (aa', b + ab'), \qquad (a, b)^{-1} = (a^{-1}, -a^{-1}b), \qquad (a, b) = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}.$$

It acts on $\mathbb{R}$ by $x \mapsto ax + b$. By Lecture 16, $dx = a^{-2}\, da\, db$ is a left Haar measure and $\Delta_G(a, b) = 1/a$. Let $N = \{n_c = (1, c)\}$ be the translations and $A = \{(a, 0)\}$ the scalings. Then $N \cong \mathbb{R}$ has Haar measure dc and $\Delta_N = 1$. For a *frequency* $\xi \in \mathbb{R}$, the map $\chi_\xi(n_c) = e^{i\xi c}$ is a unitary character of N.

17.4.1 The quotient and two measures

Since $(a,b)n_c = (a, b+ac)$, the coset $(a,b)N$ is $\{a\}\times\mathbb{R}$. So $(a,b)N \mapsto a$ identifies G/N with $(0,\infty)$, and $s(a) = (a,0)$ is a global section. The action is $(a_0,b_0)\cdot a = a_0 a$.

Since $\Delta_G(n_c) = 1 = \Delta_N(n_c)$, the transformation law (17.3) says $\rho(xn) = \rho(x)$. Thus rho-functions are positive continuous functions of a alone. The substitution $b = ac$ gives

$$F^N(a) = \int_{\mathbb{R}} F(a,ac)\,dc = \frac{1}{a}\int_{\mathbb{R}} F(a,b)\,db.$$

Hence for $\rho = \rho(a)$,

$$\int_G F\rho\,dx = \int_0^\infty \Big(\int_{\mathbb{R}} F(a,b)\,db\Big)\frac{\rho(a)\,da}{a^2} = \int_0^\infty F^N(a)\,\rho(a)\,\frac{da}{a}.$$

By the uniqueness in the quotient-measure theorem, Theorem 17.3, $d\mu_\rho = \rho(a)\,da/a$.

We use two choices. The rho-function $\rho_1 = 1$ gives the invariant measure, and the rho-function $\rho_L(a,b) = a$ gives Lebesgue measure:

$$d\mu_{\rho_1} = \frac{da}{a}, \qquad d\mu_{\rho_L} = da.$$

For the Lebesgue choice, take $g = (a_0,b_0)$ and $x = (a,0)$. Then $gx = (a_0 a, b_0)$, so $\lambda_L(g,a) = a_0$. In this case the quasi-invariance identity (17.6) is the substitution rule $\int_0^\infty \varphi(a/a_0)\,da = a_0\int_0^\infty \varphi(u)\,du$. The cocycle identity reads $a_1 a_2 = a_1\cdot a_2$, because $g_1 g_2$ has first entry $a_1 a_2$.

17.4.2 Equivariant functions and the action

For $\sigma = \chi_\xi$, the equivariance condition (17.1) says $f(a,b+ac) = e^{-i\xi c} f(a,b)$. Take $b = 0$ and then $c = b/a$:

$$f(a,b) = e^{-i\xi b/a}\varphi(a), \qquad \varphi = f\circ s. \tag{17.8}$$

Conversely, for $\varphi \in C_c((0,\infty))$ this f lies in $\mathcal{F}_c(\chi_\xi)$, since $e^{-i\xi(b+ac)/a} = e^{-i\xi c}e^{-i\xi b/a}$. So $f\mapsto\varphi$ is a bijection onto $C_c((0,\infty))$. Moreover $n_f = |\varphi|^2$ and $\|f\|^2 = \int_0^\infty |\varphi|^2\rho(a)\,da/a$.

Let $g = (a_0,b_0)$. Then $g^{-1}(a,0) = (a/a_0,\ -b_0/a_0)$, and the equivariant form (17.8) gives

$$f(g^{-1}(a,0)) = \exp\Big(-i\xi\,\frac{-b_0/a_0}{a/a_0}\Big)\varphi\Big(\frac{a}{a_0}\Big) = e^{i\xi b_0/a}\,\varphi\Big(\frac{a}{a_0}\Big).$$

The factor at $x = (a,0)$ is $(\rho(a/a_0)/\rho(a))^{1/2}$. This yields two pictures:

$$(\Pi_\xi(a_0,b_0)\varphi)(a) = e^{i\xi b_0/a}\,\varphi(a/a_0) \qquad \text{on } L^2((0,\infty),da/a), \tag{17.9}$$

$$(\Pi^L_\xi(a_0,b_0)\psi)(a) = a_0^{-1/2}e^{i\xi b_0/a}\,\psi(a/a_0) \qquad \text{on } L^2((0,\infty),da). \tag{17.10}$$

The normalised-induction theorem, Theorem 17.5, already gives unitarity. We now verify unitarity and the representation law directly in this model.

Proposition 17.9. *Let $g = (a_0,b_0)$, $g_j = (a_j,b_j)$ and $\varphi,\psi\in C_c((0,\infty))$.*

(a) $\|\Pi^L_\xi(g)\psi\| = \|\psi\|$, while $\psi\mapsto e^{i\xi b_0/a}\psi(a/a_0)$ multiplies $\|\psi\|^2$ by a_0.

(b) $\Pi^L_\xi(g_1)\Pi^L_\xi(g_2) = \Pi^L_\xi(g_1 g_2)$.

(c) $(W\varphi)(a) = a^{-1/2}\varphi(a)$ *extends to a unitary map* $L^2(da/a) \to L^2(da)$ *with* $W\Pi_\xi(g) = \Pi^L_\xi(g)W$.

Proof. (a) Substitute $a = a_0 u$:

$$\int_0^\infty a_0^{-1}|\psi(a/a_0)|^2\,da = \int_0^\infty |\psi(u)|^2\,du.$$

Without the factor a_0^{-1} the result is $a_0\|\psi\|^2$.

(b) Since $g_1 g_2 = (a_1 a_2,\ b_1 + a_1 b_2)$,

$$\begin{aligned}(\Pi^L_\xi(g_1)\Pi^L_\xi(g_2)\psi)(a) &= a_1^{-1/2} e^{i\xi b_1/a}\,(\Pi^L_\xi(g_2)\psi)(a/a_1)\\ &= a_1^{-1/2}a_2^{-1/2}\,e^{i\xi b_1/a}e^{i\xi a_1 b_2/a}\,\psi\Big(\frac{a}{a_1 a_2}\Big)\\ &= (a_1 a_2)^{-1/2} e^{i\xi(b_1+a_1 b_2)/a}\,\psi\Big(\frac{a}{a_1 a_2}\Big) = (\Pi^L_\xi(g_1 g_2)\psi)(a).\end{aligned}$$

The square roots combine because λ_L is multiplicative; this is the cocycle identity. The phases combine into the second entry of $g_1 g_2$; this is where equivariance enters.

(c) We have $\int |a^{-1/2}\varphi|^2\,da = \int |\varphi|^2\,da/a$, and W has inverse multiplication by $a^{1/2}$. Finally,

$$(\Pi^L_\xi(g)W\varphi)(a) = a_0^{-1/2}e^{i\xi b_0/a}\Big(\frac{a}{a_0}\Big)^{-1/2}\varphi\Big(\frac{a}{a_0}\Big) = a^{-1/2}e^{i\xi b_0/a}\varphi\Big(\frac{a}{a_0}\Big) = (W\Pi_\xi(g)\varphi)(a). \qquad \blacksquare$$

Note that W is multiplication by $(\rho_1/\rho_L)^{1/2}$.

17.5 The scaling orbit of a frequency

On translations, the first picture (17.9) gives $(\Pi_\xi(n_c)\varphi)(a) = e^{i(\xi/a)c}\varphi(a)$. So at the point $a \in G/N$ the group N acts by the character $\chi_{\xi/a}$. Let A act on characters of N by $(t\cdot\chi)(n) = \chi(t^{-1}nt)$. For $t = (a, 0)$,

$$t^{-1}n_c t = (a^{-1}, a^{-1}c)(a, 0) = (1, c/a), \qquad \text{so} \qquad t\cdot\chi_\xi = \chi_{\xi/a}.$$

Thus the character at aN is $s(a)\cdot\chi_\xi$. As a runs through $(0,\infty)$, the frequency ξ/a fills the orbit $\mathbb{R}_{>0}\,\xi$. There are three orbits: $(0,\infty)$, $(-\infty, 0)$ and $\{0\}$.

Proposition 17.10. *Let* $\kappa > 0$. *The operator* $(U\varphi)(a) = \varphi(a/\kappa)$ *is unitary on* $L^2((0,\infty), da/a)$, *and* $U\Pi_\xi(g) = \Pi_{\kappa\xi}(g)U$ *for all* $g \in G$.

Proof. The substitution $a = \kappa u$ preserves da/a, so U is isometric on $C_c((0,\infty))$. Its inverse is $\varphi \mapsto \varphi(\kappa\,\cdot)$. For $g = (a_0, b_0)$,

$$\begin{aligned}(U\Pi_\xi(g)\varphi)(a) &= (\Pi_\xi(g)\varphi)(a/\kappa) = e^{i\xi b_0\kappa/a}\,\varphi\Big(\frac{a}{\kappa a_0}\Big),\\ (\Pi_{\kappa\xi}(g)U\varphi)(a) &= e^{i\kappa\xi b_0/a}\,(U\varphi)(a/a_0) = e^{i\kappa\xi b_0/a}\,\varphi\Big(\frac{a}{a_0\kappa}\Big).\end{aligned}$$

The two expressions agree on the dense subspace $C_c((0,\infty))$ of $L^2((0,\infty), da/a)$. Both sides are bounded operators, so they agree on the whole space. $\blacksquare$

Thus, by Proposition 17.10, the unitary equivalence class of Π_ξ depends only on the scaling orbit of ξ. We write $\Pi_+ = \Pi_1$ and $\Pi_- = \Pi_{-1}$. By Exercise 17.5, the three representations Π_+, Π_- and Π_0 are pairwise inequivalent.

In Π_0 we have $(\Pi_0(a_0, b_0)\varphi)(a) = \varphi(a/a_0)$. So N acts trivially. The representation Π_0 is the left regular representation of $(0, \infty) \cong G/N$, composed with the quotient map $G \to G/N$. Irreducibility is not discussed here.

A quotient with no invariant Radon measure

For comparison, give the scaling subgroup $A = \{(a, 0) : a > 0\}$ the Haar measure da/a and induce its trivial character. Identify G/A with the line by $(a, b)A \mapsto b$. Left multiplication gives the familiar affine action

$$(a_0, b_0) \cdot t = b_0 + a_0 t.$$

Lebesgue measure dt is quasi-invariant, but a dilation multiplies it by a_0. Any invariant Radon measure would be translation invariant. By Haar uniqueness on $\mathbb{R}$, it would be a constant multiple of dt. Dilation invariance then forces this constant to be zero.

Here $\Delta_A = 1$ and $\Delta_G(a, 0) = 1/a$, so a rho-function is $\rho(a, b) = a$. The quotient identity follows directly:

$$\int_G F(a, b)\rho(a, b)\,\frac{da\,db}{a^2} = \int_{\mathbb{R}} \left(\int_0^\infty F(a, t)\,\frac{da}{a} \right) dt.$$

Since $(1, t)(a, 0) = (a, t)$ and A carries the Haar measure da/a, the inner integral is the coset average $F^A((1, t)A)$ of (17.2) with $H = A$. By the uniqueness in the quotient-measure theorem, Theorem 17.3, dt is exactly the measure μ_ρ supplied by this rho-function.

The square-root factor in the normalised action (17.7) becomes $a_0^{-1/2}$, and the induced action is

$$(\Pi(a_0, b_0)u)(t) = a_0^{-1/2} u\left(\frac{t - b_0}{a_0} \right).$$

To verify its norm, set $r = (t - b_0)/a_0$. Then $dt = a_0\,dr$, so

$$\|\Pi(a_0, b_0)u\|_2^2 = a_0^{-1} \int_{\mathbb{R}} |u((t - b_0)/a_0)|^2 dt = \int_{\mathbb{R}} |u(r)|^2 dr.$$

Without the square root, the squared norm would be multiplied by a_0. This quotient makes the measure correction visible in one variable. In the earlier translation-subgroup model, the quotient was the positive scaling line and an invariant quotient measure was available. The subgroup, not just the ambient group, controls the answer.

Self-checks

1. In the $ax + b$ group, derive the transformation law of a rho-function for $H = A$ and check that $\rho(a, b) = a$ satisfies it. Is $\rho = 1$ possible?
   Answer. For $h = (t, 0)$ we have $\Delta_A(h)/\Delta_G(h) = 1/(1/t) = t$, so we need $\rho(xh) = t\rho(x)$. Since $(a, b)(t, 0) = (at, b)$, the function $\rho(a, b) = a$ works. The choice $\rho = 1$ fails for $t \neq 1$.

2. Is $\rho(a,b) = a + b^2$ a rho-function for $H = N$?
   Answer. No. A rho-function for N must satisfy $\rho(xn) = \rho(x)$. But $\rho((a,0)n_1) = \rho(a,a) = a + a^2 \neq a = \rho(a,0)$.
3. Give $\Pi_\xi(g)^*$ for $g = (a_0, b_0)$ in the first picture (17.9).
   Answer. Since Π_ξ is unitary, the adjoint is $\Pi_\xi(g^{-1})$, with $g^{-1} = (a_0^{-1}, -a_0^{-1}b_0)$. Hence

$$(\Pi_\xi(g)^*\varphi)(a) = e^{-i\xi b_0/(a_0 a)}\varphi(a_0 a).$$

   Substituting $a = a_0 u$ in $\langle \Pi_\xi(g)\varphi, \psi\rangle$ confirms this formula.

Summary

A rho-function gives a quasi-invariant measure μ_ρ on G/H. Its translation factor $\lambda(g, xH) = \rho(gx)/\rho(x)$ is a cocycle. The action $f \mapsto \lambda(g^{-1}, \cdot)^{1/2} f(g^{-1}\,\cdot)$ on equivariant functions is unitary, because the cocycle cancels the change of measure. For the representation Π_ξ of the $ax + b$ group induced from the character χ_ξ of the translation subgroup N, the point $a \in G/N$ carries the frequency ξ/a. The unitary equivalence class of Π_ξ depends only on the scaling orbit of ξ.

Reading

Folland, Abstract Harmonic Analysis, Chapter 2, “Measures on Homogeneous Spaces”: quotient measures and rho-functions for a general closed subgroup. Kirillov, Elements, Section 9.1: quasi-invariant measures on homogeneous spaces. Kirillov, Elements, Section 13: models of induced representations. A. W. Knapp, *Representation Theory of Semisimple Groups* (2001), Sections VII.1–VII.2: pictures of representations induced from a parabolic subgroup of a linear connected reductive group.